Progressive block-size sweeps on DIV2K-8q

Five families × two inits × four rates ($m = n = 8$, 256×256)

Mean test PSNR (dB) over 50 images. n/a = stage undefined (mera needs k a power of 2).

1 Natural images

DIV2K natural images at 256×256. Classical block-DCT-8 is strong here (34.0 dB @ $\rho = 0.20$). Each stage k trains a $2^k \times 2^k$ inner circuit replicated by `BlockedBasis`.

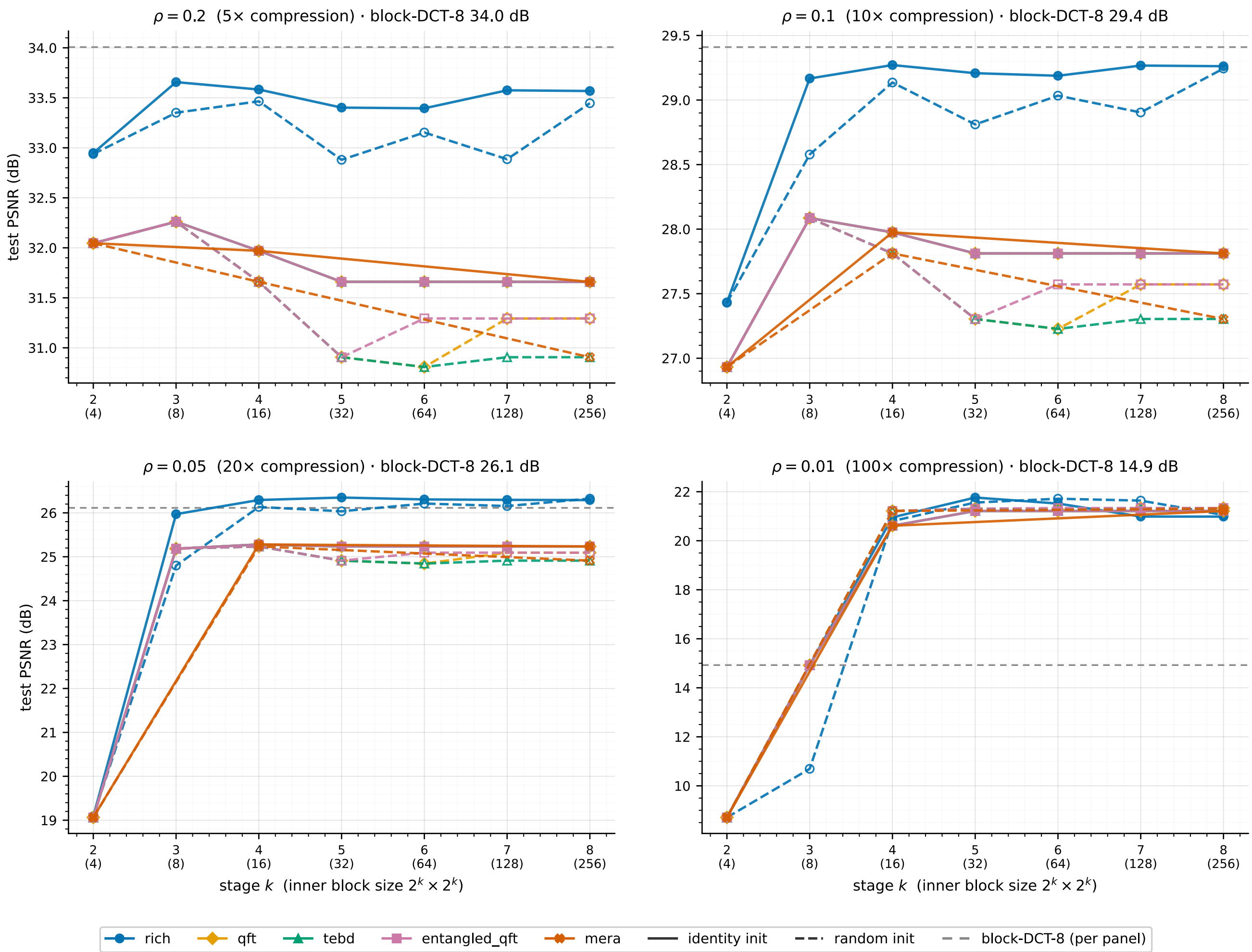

Figure 1: PSNR vs block size k at four keep ratios. One colour per family, solid = identity / dashed = random init; grey dashed = block-DCT-8 (value per panel).

2 Observations

- **`rich` leads the learned families** ($\approx$ 33.7 dB @ $\rho = 0.20$, $\approx$ 1.5 dB above the rest) but sits just below block-DCT-8 (34.0).
- **The learned circuits overtake block-DCT as compression tightens** — `rich` draws level by $\rho = 0.05$ and every family clears it by $\approx$ 6–7 dB at $\rho = 0.01$.
- **Under identity init the four QFT-derived families are bit-identical**; identity $\geq$ random; curve flat in k.

3 Per-rate tables (mean PSNR, dB)

$\rho = 0.20$ — **5×**

family	init	k=2\ (4)	k=3\ (8)	k=4\ (16)	k=5\ (32)	k=6\ (64)	k=7\ (128)	k=8\ (256)
rich	identity	32.9	33.7	33.6	33.4	33.4	33.6	33.6
rich	random	32.9	33.4	33.5	32.9	33.2	32.9	33.4
qft	identity	32	32.3	32	31.7	31.7	31.7	31.7

family	init	k=2\ (4)	k=3\ (8)	k=4\ (16)	k=5\ (32)	k=6\ (64)	k=7\ (128)	k=8\ (256)
qft	random	32	32.3	31.7	30.9	30.8	31.3	31.3
tebd	identity	32	32.3	32	31.7	31.7	31.7	31.7
tebd	random	32	32.3	31.7	30.9	30.8	30.9	30.9
ent-qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
ent-qft	random	32	32.3	31.7	30.9	31.3	31.3	31.3
mera	identity	32	n/a	32	n/a	n/a	n/a	31.7
mera	random	32	n/a	31.7	n/a	n/a	n/a	30.9

$\rho = 0.10$ — **10**×

family	init	k=2\ (4)	k=3\ (8)	k=4\ (16)	k=5\ (32)	k=6\ (64)	k=7\ (128)	k=8\ (256)
rich	identity	27.4	29.2	29.3	29.2	29.2	29.3	29.3
rich	random	27.4	28.6	29.1	28.8	29	28.9	29.2
qft	identity	26.9	28.1	28	27.8	27.8	27.8	27.8
qft	random	26.9	28.1	27.8	27.3	27.2	27.6	27.6
tebd	identity	26.9	28.1	28	27.8	27.8	27.8	27.8
tebd	random	26.9	28.1	27.8	27.3	27.2	27.3	27.3
ent-qft	identity	26.9	28.1	28	27.8	27.8	27.8	27.8
ent-qft	random	26.9	28.1	27.8	27.3	27.6	27.6	27.6
mera	identity	26.9	n/a	28	n/a	n/a	n/a	27.8
mera	random	26.9	n/a	27.8	n/a	n/a	n/a	27.3

$\rho = 0.05$ — **20**×

family	init	k=2\ (4)	k=3\ (8)	k=4\ (16)	k=5\ (32)	k=6\ (64)	k=7\ (128)	k=8\ (256)
rich	identity	19.1	26	26.3	26.3	26.3	26.3	26.3
rich	random	19.1	24.8	26.1	26	26.2	26.2	26.3
qft	identity	19.1	25.2	25.3	25.2	25.2	25.2	25.2
qft	random	19.1	25.2	25.2	24.9	24.8	25.1	25.1
tebd	identity	19.1	25.2	25.3	25.2	25.2	25.2	25.2
tebd	random	19.1	25.2	25.2	24.9	24.8	24.9	24.9
ent-qft	identity	19.1	25.2	25.3	25.2	25.2	25.2	25.2
ent-qft	random	19.1	25.2	25.2	24.9	25.1	25.1	25.1
mera	identity	19.1	n/a	25.3	n/a	n/a	n/a	25.2
mera	random	19.1	n/a	25.2	n/a	n/a	n/a	24.9

$\rho = 0.01$ — **100**×

family	init	k=2\ (4)	k=3\ (8)	k=4\ (16)	k=5\ (32)	k=6\ (64)	k=7\ (128)	k=8\ (256)
rich	identity	8.7	14.9	21	21.8	21.5	21	21
rich	random	8.7	10.7	20.8	21.6	21.7	21.6	21.1
qft	identity	8.7	14.9	20.6	21.2	21.2	21.2	21.2
qft	random	8.7	14.9	21.2	21.3	21.3	21.3	21.3
tebd	identity	8.7	14.9	20.6	21.2	21.2	21.2	21.2
tebd	random	8.7	14.9	21.2	21.3	21.3	21.3	21.3
ent-qft	identity	8.7	14.9	20.6	21.2	21.2	21.2	21.2
ent-qft	random	8.7	14.9	21.2	21.3	21.3	21.3	21.3
mera	identity	8.7	n/a	20.6	n/a	n/a	n/a	21.2
mera	random	8.7	n/a	21.2	n/a	n/a	n/a	21.3